Additive Manufacturing 59 (2022) 103089

Contents lists available at ScienceDirect

Additive Manufacturing

journal homepage: www.elsevier.com/locate/addma

Machine-learning assisted optimization of process parameters for controlling the microstructure in a laser powder bed fused WC/Co cemented carbide

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ARTICLE INFO

Keywords:
Laser powder bed fusion
Composite
Machine learning
Optimization
Moving heat source

ABSTRACT

A Single laser scan was performed on a sintered WC/Co composite and the process conditions were optimized to realize a WC/Co two-phase microstructure in a WC/Co composite fabricated by laser powder bed fusion (L-PBF). Laser irradiation on the sintered WC/Co composites generated two different microstructural regions. One is composed of WC and Co phases (WC/Co two-phase region), and the other includes W_2C and W_3Co_3C phases (WC decomposition region). In the L-PBF-fabricated WC/Co composites, these two microstructural regions are distributed in a complex manner and their fractions varies depending on the laser parameters. Using a convolutional neural network model that is trained with the micrographs of single-laser-scanned WC/Co composites, the fractions of the two microstructural regions in the WC/Co composites fabricated by the L-PBF under various laser conditions were quantified. Multiple linear regression analyses revealed that the laser power and spot size contribute more than the scan speed, which has been verified by numerical analysis based on the moving heat source model. A support vector machine (SVM) suggested that low laser power and low scan speed are required for suppressing the WC decomposition in the consolidated samples. The WC decomposition has been successfully suppressed in the samples fabricated under SVM-recommended conditions.

1. Introduction

The recent developments in additive manufacturing (AM) have made it possible to manufacture complex-shaped metal parts [1–4]. The applications of metal AM technologies have been expanded to various fields such as the medical, aerospace, and automobile industries [5–7]. Laser powder bed fusion (L-PBF) is one of the most widely used metal AM technologies. Among other AM technologies, L-PBF is characterized as a process by which complex-shaped products are manufactured more precisely [2].

WC/Co composite products are widely used as cutting tools and molds that require high hardness and wear resistance. They are usually manufactured by sintering raw powders and cutting the sintered products into desired shapes [8]. During the production of WC/Co composite products, long manufacturing time, poor material yield, and limitations to fabricate complex shapes are some of the drawbacks [9]. This problem can be solved by using L-PBF technologies for producing WC/Co composite products. Products can be manufactured in a near-net shape, resulting in shorter processing times and improved material yields. In addition, a complex geometry including an internal cooling structure is formed by the L-PBF process. The cooling structure improves the life of tools and molds.

The L-PBF technologies have been applied for fabricating the WC/Co composite parts. [10,11]. A relative density of 95–98% is achieved under optimized laser powers and scan speeds. However, the manufactured products endure numerous cracks which caused the deterioration of mechanical properties [12,13]. The WC particles in the L-PBF-fabricated WC/Co composite samples are coarsened, and the mechanical properties deteriorated [14]. Therefore, it is difficult to fabricate the WC/Co composites by the L-PBF process with appropriate mechanical strength for practical applications [15–20]. Heat treatments at temperatures above the melting temperature of Co are effective in mitigating the defects (pores and cracks) [21]. The molten Co flows into the defects and increases the relative density. The hot isostatic press (HIP) treatment is also effective in eliminating defects and increasing the relative density [22].

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https://doi.org/10.1016/j.addma.2022.103089
Received 4 May 2022; Received in revised form 5 August 2022; Accepted 12 August 2022
Available online 17 August 2022

It has been reported that L-PBF-fabricated WC/Co composites are composed of two microstructural regions [21,24]. One region consists of a two-phase WC/Co microstructure (denoted as WC/Co two-phase region herein), and the other region is composed of numerous defects and undesirable phases (W_2C and W_3Co_3C phases) (denoted as the WC decomposition region herein). The WC decomposition region is peculiar to L-PBF-fabricated WC/Co composites and has a detrimental effect on their mechanical properties. The WC phase is decomposed and melted closer to the laser irradiation spot. The volatilization of Co and C during the melting process causes the chemical composition to shift in a W-rich direction, and the subsequent non-equilibrium solidification results in the formation of W_2C and W_3Co_3C phases. It has been reported that post-heat treatment is effective in eliminating these phases. The microstructure and constituent phases are changed by post-heat treating the samples at elevated temperatures [23]. However, the samples heat-treated below 1000 °C still contain the W_3Co_3C, $W_9Co_3C_4$, and W_6Co_6C phases. The HIP treatment also changes the microstructure of L-PBF-fabricated WC/Co composites [22]. However, W_3Co_3C and W_6Co_6C phases are present even after the HIP treatment [22]. Recently, Ibe et al. revealed that the shift in composition during the L-PBF process stabilizes the brittle phases at heat treatment temperature [21]. In anticipation of the composition shift, the carbon-added WC/Co composite powder was processed by the L-PBF and subsequently heat-treated above the liquidus temperature of the Co phase, which results in successfully forming a microstructure of WC and Co phases [21]. Therefore, post-heat treatment is an effective approach for eliminating the brittle phases and tailoring the microstructure.

However, the Vickers hardness of the heat-treated WC/Co composites fabricated by the L-PBF process is lower than that of conventionally sintered WC/Co composites [21]. This is caused by the significant coarsening of the WC phase through L-PBF and heat treatment. High-energy laser irradiation causes coarsening of the WC phase in the WC/Co two-phase region in the L-PBF-fabricated WC/Co composites. In addition, the transformations from W_2C and W_3Co_3C phases to the WC phase by heat treatment result in the formation of a coarsened WC phase. Therefore, to fabricate high-strength WC/Co composites using the L-PBF process, it is important to optimize the laser parameters and control the microstructure of the as-built WC/Co composites.

Recently, machine-learning approaches have been rapidly developed and applied for the optimization process of AM technologies. Tapia et al. created a surrogate model for predicting melt-pool depth and porosity using Gaussian process (GP) regression [25,26]. Aoyagi et al. optimized the process parameters of electron beam powder bed fusion using a support vector machine (SVM) [27]. They classified the surface state of 11 samples into two categories (good or bad) and created the map of the decision function, which corresponds to the distance from the hyperplane separating good and bad. Appropriate current and scan speed were estimated by selecting the condition located furthest from the hyperplane (the condition with the largest decision function), and a dense sample was successfully manufactured under the condition. Liu et al. created a process map for an L-PBF-fabricated AlSi10Mg alloy using GP regression [28]. The GP regression identified a wider process window compared with the conventionally reported process window. New indices for describing the cellular microstructure of the L-PBF-fabricated AlSi10Mg alloy were also developed. Yanase et al. investigated the densification conditions for L-PBF-fabricated AlSi10Mg alloy using a neural network, principal component analysis, and hierarchical clustering [29]. It was revealed that the contribution of the laser parameters to the densification process changed with respect to energy density.

Thus, machine learning is effective for exploring the process conditions to fabricate high-density materials using PBF technologies. However, there is limited research on the optimization of laser parameters to control the microstructure with the assistance of machine learning. Achieving microstructural control assisted by machine learning leads to the development of L-PBF-fabricated materials with high functionality. In the present study, the laser parameters for suppressing the formation of the WC decomposition region in an L-PBF-fabricated WC/Co composite were explored using machine learning integrated with a single laser scanning experiment. WC/Co composite samples were fabricated using the L-PBF process under various laser conditions. A convolutional neural network (CNN) was utilized for the segmentation of the microstructure of the L-PBF-fabricated WC/Co composites. To reduce the cost of preparing the training data, micrographs of single-laser-scanned WC/Co composites were used for the training data of the CNN. The area fraction of the WC decomposition region (A_d) was quantified using the CNN. The relationship between A_d and laser conditions was analyzed using multiple linear regression (MLR) and SVM. These results were used for discussing the contribution of each laser parameter to the formation and increase of the WC decomposition region.

2. Materials and methods

2.1. Single laser scanning experiment

A sintered WC/20 mass% Co block (VU70, Silver Alloy Co., Ltd, Japan) of dimensions 20 × 80 × 10 mm was used for the single-laser scanning experiments. The built parts (consolidated parts) are below the powder bed during the L-PBF process, and therefore the thermal properties (e.g., thermal conductivity) of the laser-scanned region are presumably close to that of a block rather than that of powder. Therefore, a block sample was used instead of powder. A laser scan apparatus (K-MFSC, Kisoh, Japan) equipped with an Yb fiber laser (maximum laser power of 500 W and wavelength of 1070 nm) was used. Fig. 1(a) shows the laser parameters of the single-laser scanning experiment. The laser spot size was fixed as ~ 200 μm. The laser power (P) and scan speed (v) were controlled in the range of 100–500 W and 10–100 mm·s^{-1}, respectively. The laser was scanned on a surface of 20 × 80 mm^2 in the longitudinal direction at a distance of 70 mm. When the laser parameters represented by the cross symbols in Fig. 1(a) were applied, the block did not melt because of the low-energy input. Under the conditions of

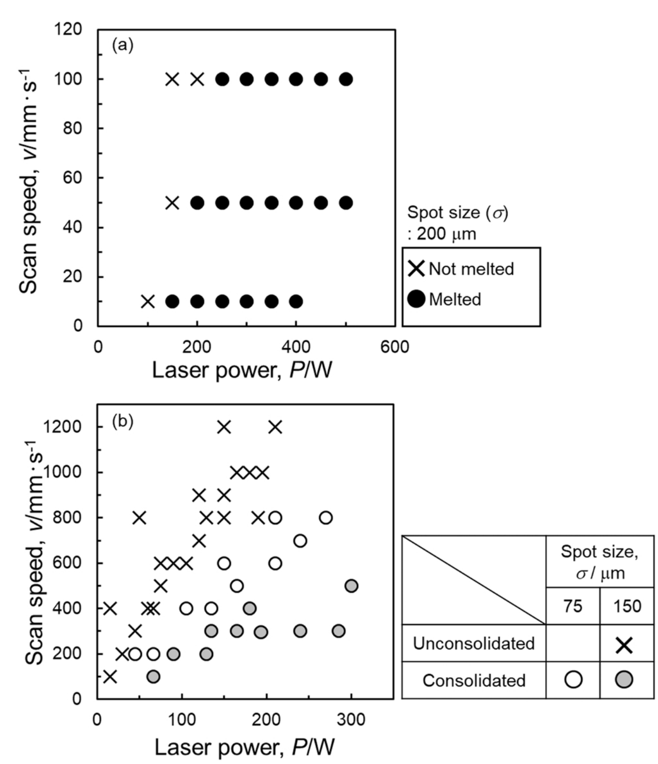

Fig. 1. Laser conditions of (a) single laser scanning on sintered WC/Co composite and (b) laser powder bed fusion of WC/Co composite samples.

laser represented by circles in Fig. 1(a), the following analyses were made.

The laser-scanned specimens were cut perpendicular to the direction of the scan. The cross section was mechanically polished using a plate with diamond abrasive grains (#500), buffed with diamond slurries of 9 μm and 3 μm, and finished using colloidal silica. The microstructure was observed using scanning electron microscopy (SEM; JSM-6610A, JEOL, Japan). The chemical compositions of the phases in the sintered WC/Co composites that were laser-scanned under the condition of (P, v) = (300 W, 100 mm·s^{-1}) were analyzed by energy-dispersive X-ray spectroscopy (EDS). The constituent phases of the sintered WC/Co composites that were laser-scanned with the parameters of (P, v) = (300 W, 100 mm·s^{-1}) and (500 W, 10 mm·s^{-1}) were investigated using an X-ray diffraction (XRD) with a Cu radiation source. The voltage and current were maintained at 40 kV and 40 mA, respectively. The depth and width of the WC decomposition region and WC/Co two-phase region were measured using an image analysis software (Image J, National Institutes of Health, USA).

2.2. Laser powder bed fusion process

A WC/25 mass% Co composite powder was fabricated by granulation and sintering. The detailed method of fabricating the powders is described elsewhere [21,24]. The powder had a median diameter (D50) of approximately 18 μm and WC particle size of approximately 1.5 μm. The laser powder bed fusion process was performed using a 3D printer

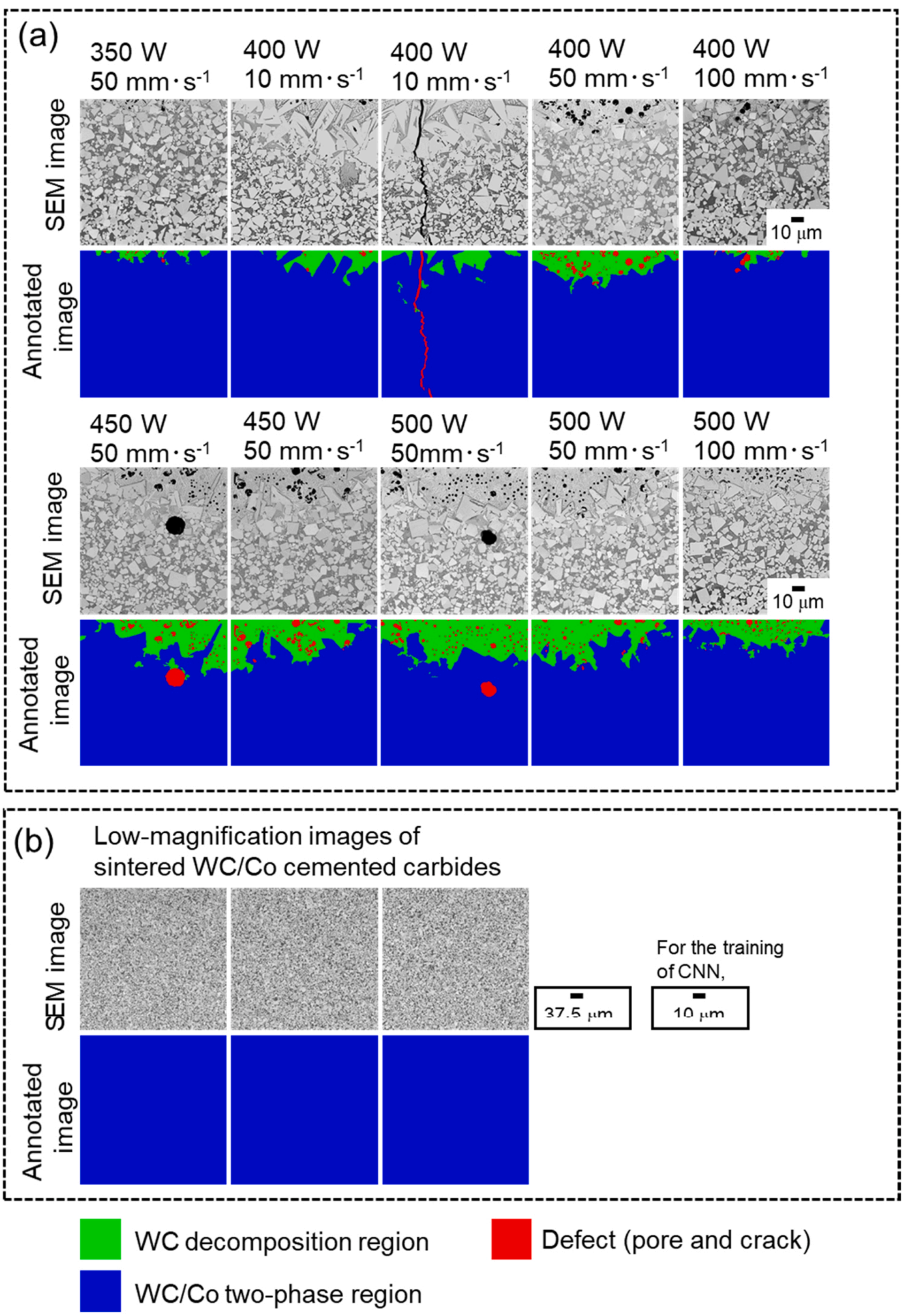

Fig. 2. (a) SEM images and corresponding manually annotated images of sintered WC/Co composites that were single-laser-scanned under various laser conditions. The SEM and annotated images were used for training characteristics of each region to a convolutional neural network (CNN). (b) Low-magnification SEM images and corresponding manually annotated images of sintered WC/Co composites. The SEM and annotated images were used for training WC/Co two-phase microstructure containing fine WC particles. The SEM images shown in (b) are at lower magnification compared to the SEM images shown in (a); however, the SEM and annotated images were trained to the CNN model as the SEM and annotated images with the same magnification as (a).

(ProX 200, 3D Systems, Rock Hill, SC, USA) under a high-purity Ar gas atmosphere with oxygen concentration of below 500 ppm. Cubic samples of dimensions 10 × 10 × 10 mm^3 were fabricated under various laser conditions such as input powers, scan speeds, and spot sizes (σ), as shown in Fig. 1(b). The spot size was set as 75 μm (just focused) and 150 μm (−6 mm under focused), and the hatch spacing and powder layer thickness were fixed as 100 μm and 30 μm, respectively. A hexagonal laser scanning pattern was applied in this study. When the laser conditions represented by cross (×) in Fig. 1(b) were applied, the samples were unconsolidated owing to the low-energy input. The samples fabricated under the laser conditions represented by circle (○) in Fig. 1 (b) were cut horizontally in the building direction and polished using the same method described above. The microstructures of the samples were observed using SEM.

To investigate the mechanical properties, specimens with dimensions of 11 × 33 × 8 mm^3 were fabricated under the laser conditions of (P, v, σ) = (66 W, 100 $mm{\cdot}s^{-1}$, 150 μm) and (195 W, 300 $mm{\cdot}s^{-1}$, 150 μm). The Co content of these samples was analyzed by X-ray fluorescence spectrometer XRF-1800 (Shimadzu Corp., Kyoto, Japan). The C content was also measured by infrared absorption method using an EMIA 321V2 (Horiba, Kyoto, Japan). The as-built WC/25 mass % Co composites were heat-treated at 1380 °C for 4 h and hot-isostatic pressed at 1350 °C for 4 h under a pressure of 100 MPa to eliminate pores and cracks. The transverse rupture strength was measured on a sample of dimension 8 × 24 × 4 mm^3. The strengths of the specimens under each building condition were measured in three replicates and averaged. Vickers hardness (HV) was also evaluated using an indenter under a controlled load of 9.8 N at ambient temperature.

2.3. Data science approaches

TernausNet [30] was used as a CNN model in the present study. TernausNet is based on U-Net [31], which is a precise semantic segmentation model. Fig. 2 shows the training data for the CNN model. Fig. 2(a) presents the SEM images and corresponding manually annotated images of the sintered WC/Co composites that were single-laser-scanned under various conditions. The WC decomposition region can be observed in the upper regions of the SEM images. Pores and cracks are also observed in the figures. The SEM images were manually segmented into three regions using an image analysis software (Image J, National Institutes of Health, USA): the WC decomposition region, WC/Co two-phase region, and defects. The WC particle size in the sintered WC/Co composites is approximately 10 μm, which is significantly larger than the WC particle size (~1.5 μm) in the WC/25 mass% Co powder used for the L-PBF process [21,24]. This difference in size led to misrecognizing the WC/Co two-phase region as the WC decomposition region because the WC decomposition region has a finer microstructural pattern than the WC/Co two-phase region. Therefore, low-magnification SEM images of the sintered WC/Co composite shown in Fig. 2(b) were used as the training data. The SEM images in Fig. 2(b) were taken at a magnification of 3.75 times lower than those in Fig. 2(a) but were trained to the CNN as the images with the same magnification as in Fig. 2(a). These data play a role in training the microstructural region including fine WC particles as the WC/Co two-phase region to the CNN. Then, the regions with WC particles of several micrometers size were trained as the WC/Co two-phase microstructure. Additionally, data augmentation was performed. The data augmentations used were horizontal flip, vertical flip, random rotation (0°, 90°, 180, and 270º), and change in brightness and contrast. The probabilities of these data augmentations as well as the epoch number, batch size, and learning rate were set as hyperparameters of the CNN model. These hyperparameters were optimized via Bayesian optimization using a framework called Optuna [32]. The search ranges of the Bayesian optimization and optimized parameters are shown in Table 1. To validate the data of Bayesian optimization, the SEM images of L-PBF-fabricated WC/Co-composites fabricated under the laser conditions of (P, v, σ) = (285 W, 300 $mm{\cdot}s^{-1}$, 150 μm), (90 W, 200 $mm{\cdot}s^{-1}$, 150 μm), and (45 W, 200 $mm{\cdot}s^{-1}$, 75 μm) were manually annotated and used. Accuracy and recall were used as the precise prediction performance indices of the CNN. The accuracy and recall parameters were determined using Eqs. (1) and (2), respectively.

Table 1
Bayesian optimized hyperparameters of the convolutional neural network for semantic segmentation.

Hyperparameters		Search space	Optimized parameters	
			Training data:Fig. 2 (a)	Training data:Fig. 2 (a, b)
Epoch		0–5000 (interval: 10)	1040	4010
Batch size		1–13	2	4
Learning rate		10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}	10^{-3}	10^{-3}
Probability of data augmentation	Horizontal flip	0, 0.5	0.5	0.5
	Vertical flip	0, 0.5	0	0.5
	Random rotate 0º, 90º, 180º, and 270º	0, 1	1	0
	Change brightness & contrast	0, 1	0	1

$$\text{Accuracy} = \frac{N_{\text{true}}}{N_{\text{total}}}, \tag{1}$$

$$\text{Recall}(i) = \frac{N_{i,\text{true}}}{N_i}, \tag{2}$$

where N_{true} and N_{total} are the pixel numbers of the accurately predicted regions and overall image, respectively. N_i and $N_{i,\text{ true}}$ are the pixel numbers of class i (i = WC decomposition region, WC/Co two-phase region, and defects) and accurately predicted region as i, respectively. Using the trained CNN model, all the SEM images of the L-PBF-fabricated WC/Co composites were segmented into three classes, and the area fraction of the WC decomposition region (A_d) was calculated as follows:

$$A_d = \frac{N_{\text{WCdec}}}{N_{\text{WCdec}} + N_{\text{WC/Co}}}, \tag{3}$$

where N_{WCdec} and $N_{\text{WC/Co}}$ are the pixel numbers of the WC decomposition and WC/Co two-phase regions, respectively.

To investigate the contribution of each laser parameter to A_d, the relationship between A_d and laser conditions was analyzed using multiple linear regression (MLR). The regression equation is expressed as follows:

$$A'_d = aP' + bv' + c\sigma' + d, \tag{4}$$

A'_d, P', v', and σ' are the area fraction of the WC decomposition region, laser power, scan speed, and spot size that are standardized using the average and standard deviation of the dataset, respectively. a, b, and c are standardized partial regression coefficients that represent the contribution of each laser parameter. d is the intercept and is nearly zero when standardization was carried out.

A support vector machine (SVM) was used to explore the laser conditions for suppressing the formation of WC decomposition region. SVM enables the classification of laser parameters by finding the hyperplane with the largest distance from the data points. In the present study, two criteria were set for the classification: consolidated/unconsolidated (denoted by ○ or × in Fig. 1(b)) and A_d< 0.1/A_d≧ 0.1. The first criterion was set to explore the manufacturable laser conditions, whereas the second criterion was set to explore the laser conditions for suppressing the formation of the WC decomposition region. A linear kernel was applied in this study. The regularization coefficient was set to 1.0.

3. Results

3.1. Microstructure in single-laser-scanned WC/Co composites

Fig. 3(a–c) shows the representative microstructure of the sintered WC/Co composite that was single-laser scanned under a laser power of 500 W and a scan speed of 100 mm•s^{-1}. The top surface on which the laser beam was irradiated is shown in Fig. 3(a). The microstructures were roughly classified into three regions. In the region closest to the laser-irradiated surface (Fig. 3(b)), a dendritic microstructure has been observed, whereas no angular WC particles have been observed. Numerous pores were formed in this region. The microstructural characteristics are similar to those of the WC decomposition region in L-PBF-fabricated WC/Co composites [21,24]. The dendritic microstructure indicated that the WC particles were melted by laser scanning. Below the WC decomposition region, WC particles were observed, which is similar to the microstructure of conventionally sintered WC/Co composites (WC/Co two-phase region) (Fig. 3(b)). However, as shown in Fig. 3(c), the Co phase in the WC/Co two-phase region had a brighter contrast compared with the Co phase in the region far away from the laser-irradiated surface (denoted as the as-sintered microstructure region). In addition, a network microstructure was formed in the Co phase (indicated by white arrows in Fig. 3(c)). The WC particles in the WC/Co two-phase region were spherical to a certain extent compared with that of WC particles in the as-sintered microstructure region (indicated by black arrows in Fig. 3(c)). Fig. 3(d, e) shows the SEM images and corresponding EDS elemental maps of C, Co, and W at the boundaries (d) between the WC decomposition and WC/Co two-phase regions and (e) between the WC/Co two-phase and as-sintered microstructure regions. W and C are highly concentrated in the phase with dendritic morphology in the WC decomposition region (Fig. 3(d)). The dendritic morphological phase has a slightly brighter contrast than the WC phase, indicating that the phase would be W_2C phase. As shown in Fig. 3(e), W was detected in the network microstructure in the WC/Co two-phase region. The spherical WC particles and EDS results indicate the dissolution of W and C into Co melt in the WC/Co two-phase region. The Co phase is melted by the heat transfer from the laser-irradiated spot. According to the phase diagram of WC-Co pseudo-binary system [33], there is a eutectic reaction (Co-rich liquid phase → Co phase + WC phase). W and

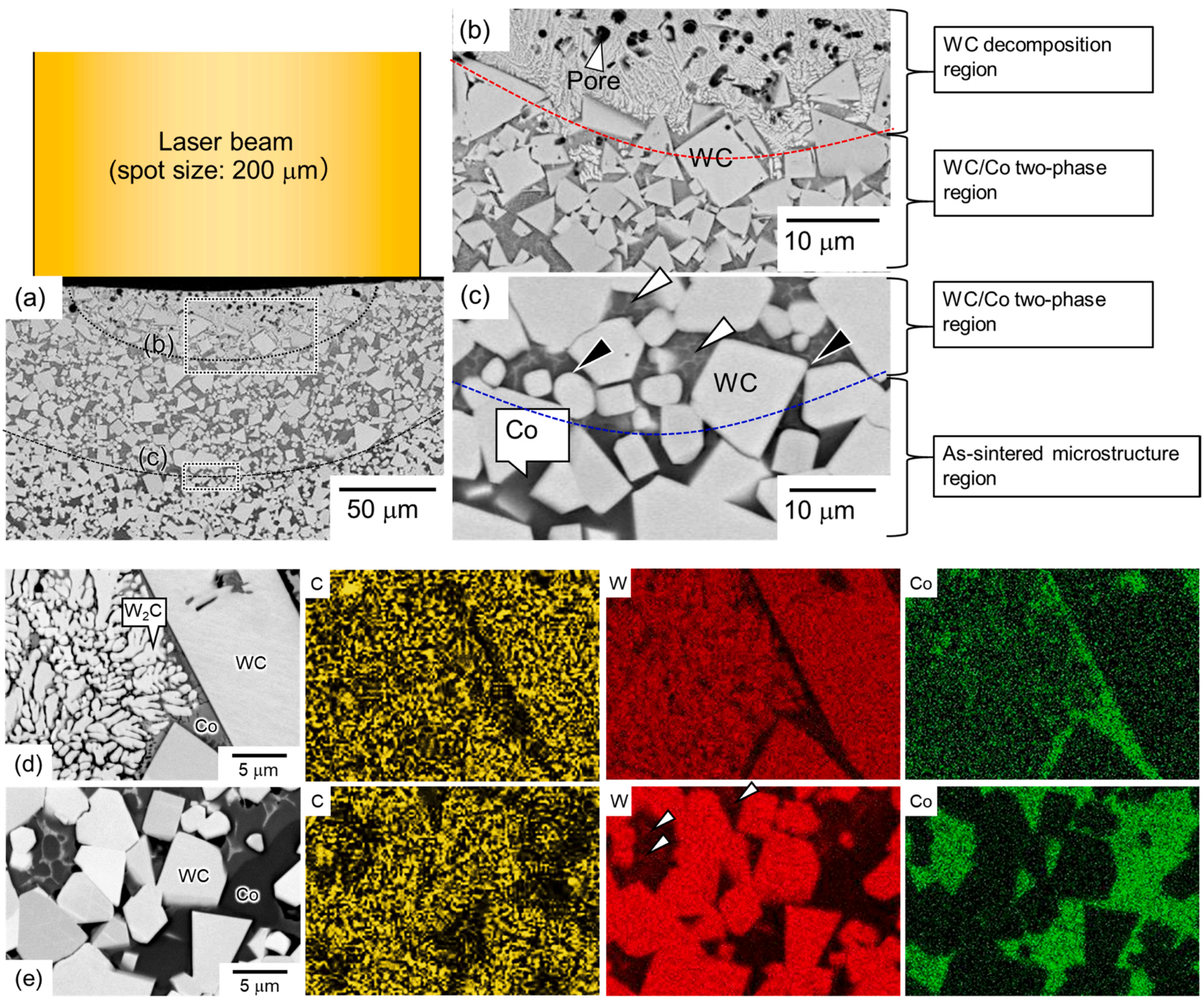

Fig. 3. (a) Low and (b, c) high-magnification SEM images showing the microstructure of sintered WC/Co composites that are single-laser-scanned under a laser power of 500 W and a scan speed of 100 mm·s^{-1}. (d, e) SEM images and corresponding EDS elemental maps of C, Co, and W at the boundaries (d) between the WC decomposition and WC/Co two-phase regions and (e) between the WC/Co two-phase and as-sintered microstructure regions.

C are dissolved into the Co-rich melt and the eutectic region is formed during solidification; therefore, resulting in the formation of network microstructure, as shown in Fig. 3(e).

Fig. 4(a) presents the SEM micrographs of sintered WC/Co composites that are single-laser-scanned under the following conditions: (P, v) = (a) (500 W, 100 mm•s^{-1}), (b) (500 W, 50 mm•s^{-1}), (c) (300 W, 100 mm•s^{-1}), and (d) (300 W, 50 mm•s^{-1}). The area energy density (E_d^A) determined using Eq. (5) is shown at the lower left part of the micrographs in Fig. 4.

$$E_d^A = \frac{P}{v \cdot \sigma}, \quad (5)$$

where E_d^A indicates the input energy per unit area of a single laser-scanned track and increases with increasing P and decreasing v. When the laser power were 500 W, the WC decomposition region was increased by lowering the scan speed from 100 mm·s^{-1} to 50 mm·s^{-1} (E_d^A = 25 J·mm^{-2} → 50 J·mm^{-2}) owing to the high energy input (Fig. 4 (a) and (b)). When the laser power was decreased from 500 W to 300 W under a scan speed of 100 mm·s^{-1} (E_d^A = 25 J·mm^{-2} → 15 J·mm^{-2}), a WC decomposition region was not formed (Fig. 4(a, c)). Even when the scan speed was decreased from 100 mm·s^{-1} to 50 mm·s^{-1} under a laser power of 300 W (E_d^A = 15 J·mm^{-2} → 30 J·mm^{-2}), a WC decomposition region was not formed (Fig. 4(c, d)). The area energy density shown in Fig. 4(d) (E_d^A = 30 J·mm^{-2}) is higher than that in Fig. 4(a) (E_d^A = 25 J·mm^{-2}). However, the WC decomposition region has not been observed in Fig. 4(d), whereas the WC decomposition region has been observed in Fig. 4(a). These results indicate that the formation of the WC decomposition region cannot be determined only by the area energy density. The XRD profiles of the sintered WC/Co composites that were single-laser-scanned under laser conditions of (P, v) = (500 W, 100 mm•s^{-1}) and (300 W, 100 mm•s^{-1}) are shown in Fig. 4(e). In the WC/Co composites that were single-laser-scanned under the conditions of (P, v) = (500 W, 100 mm•s^{-1}), peaks of W_2C and W_3Co_3C phases were detected. These phases are also observed in L-PBF-fabricated WC/Co composites [21,24], which indicates that the single-laser-scanned sample had the same constituent phases as the L-PBF-fabricated sample. The relatively high-intensity peak of the W_2C phase supports that the dendritic morphological phase in Fig. 3(d) was the W_2C phase. By contrast, only the peaks of WC and Co (fcc) phases were detected in the laser-scanned sample at the condition of (P, v) = (300 W, 100 mm•s^{-1}). Therefore, it was confirmed that the W_2C and W_3Co_3C phases were present only in the WC decomposition region.

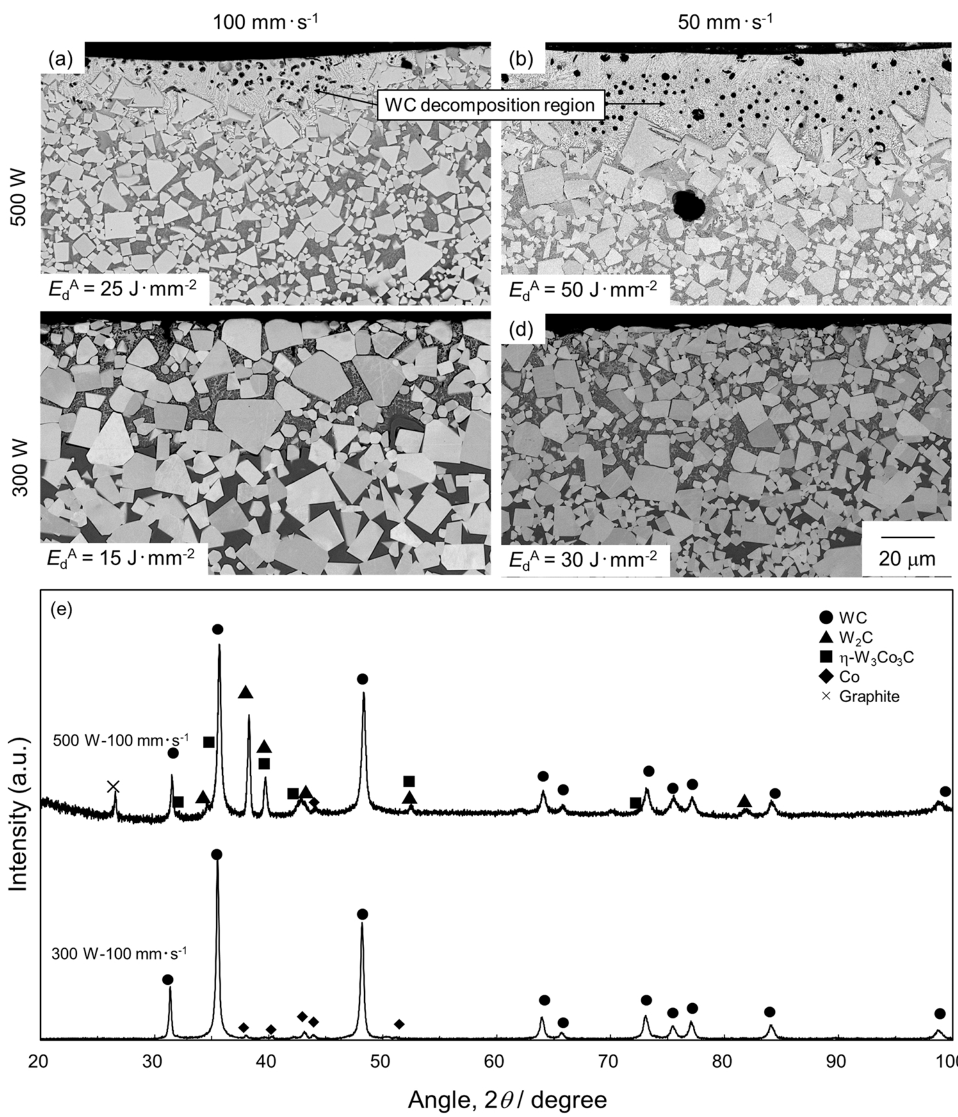

Fig. 4. SEM images of sintered WC/Co composites that are single-laser-scanned under laser powers of (a, b) 500 W and (c, d) 300 W at a scan speeds of (a, c) 100 mm·s^{-1} and (b, d) 50 mm·s^{-1}. (e) XRD profiles of sintered WC/Co composites that are single-laser-scanned under laser powers of 500 W and 300 W at a scan speed of 100 mm·s^{-1}.

Fig. 5 presents the bubble chart of the (a) width and (b) depth of the WC decomposition region and WC decomposition region + WC/Co two-phase region plotted as functions of the laser power and scan speed. The diameters of the circles indicate the width and depth of each region. The open symbols represent the total width/depth of the WC decomposition region and WC/Co two-phase microstructure region and the closed symbols represent the width/depth of the WC decomposition region. The absence of the closed symbols inside the open symbols indicates that the WC decomposition region was not formed. Both depth and width increased with increasing laser power and decreasing scan speed. The broken straight line represents the laser conditions with an E_d^A of 10 J•mm^{-2}. Along this line, the WC decomposition region was not formed under the conditions of (P, v) = (250 W, 50 mm•s^{-2}), whereas the WC decomposition region with a width of approximately 300 μm and depth of approximately 50 μm was formed under the conditions of (P, v) = (500 W, 100 mm•s^{-1}). This indicates that the formation of the WC decomposition region can be suppressed by applying a low laser power and scan speed under the same energy density. The curved broken line was roughly drawn as the boundary between the conditions for the formation and non-formation of the WC decomposition region, as shown in Fig. 5. The curve had a downward convex shape, indicating that the laser power contributed significantly to the formation of the WC decomposition region compared to the scan speed.

3.2. Data science analyses for L-PBF-fabricated WC/Co composites

Fig. 6 shows the SEM, manually annotated, and CNN annotated images with the accuracy and recall values of the L-PBF-fabricated WC/Co-composites fabricated under various laser conditions when only Fig. 2(a) was trained to the CNN. Under the conditions of (P, v, σ) = (285 W, 300 mm•s^{-1}, and 150 μm), a significant WC decomposition region was observed in the samples. However, under the conditions of (P, v, σ) = (90 W, 200 mm•s^{-1}, and 150 μm), the WC decomposition region was formed only to a small extent. Three regions could be segmented precisely in the image of the sample including the wide WC decomposition region ((P, v, σ) = (285 W, 300 mm·s^{-1}, and 150 μm)). The accuracy and recall of the pore and WC decomposition region exceeded 0.94, which indicates highly precise segmentation. However, precise segmentation could not be achieved in the images of the samples with small WC decomposition regions ((P, v, σ) = (90 W, 200 mm·s^{-1}, and 150 μm) and (45 W, 200 mm·s^{-1}, and 75 μm)). Especially, the WC/Co two-phase region tends to be misrecognized as the WC decomposition region. This would be because the CNN emphasized the size of the bright phase (WC and W_2C phase) in the classification rather than the morphology. As shown in Fig. 2(a), the microstructure in the WC decomposition region had a much finer pattern than that in the WC/Co two-phase microstructure. Therefore, the CNN recognizes the fine microstructural regions as the WC decomposition, resulting in misclassifying the WC/Co two-phase region with finer WC particles in the L-PBF-fabricated WC/Co composites as the WC decomposition region.

Fig. 7 shows the SEM, manually annotated, and CNN annotated images with the accuracy and recall values of the L-PBF-fabricated WC/Co-composites fabricated under various laser conditions when both Fig. 2(a) and (b) were trained to the CNN. The accuracy value was improved compared with Fig. 6. The CNN annotated WC decomposition regions were relatively close to manual annotation in view of the location and morphology. The relatively accurate segmentation would be caused by adding Fig. 2(b) as training data. The CNN would learn the microstructural region with fine WC particles (low magnified WC particles) as the WC/Co two-phase region. Then, the CNN model would classify the microstructural region based on the morphology of the bright phase rather than the size. Using the CNN model, all 20 SEM images of the L-PBF-fabricated WC/Co composites were annotated, and the area fraction of the WC decomposition region (A_d) was analyzed using Eq. (3). The CNN annotation took approximately 15 s for all 20 images (Intel(R) Xeon(R) W-2123 CPU @ 3.60 GHz, computer memory: 32 GB, and graphics card: NVIDIA GeForce RTX2080 SUPER), which is much faster than manual annotation.

Fig. 8 presents (a) the bubble chart of A_d plotted as functions of laser power and scan speed and (b) representative SEM and corresponding

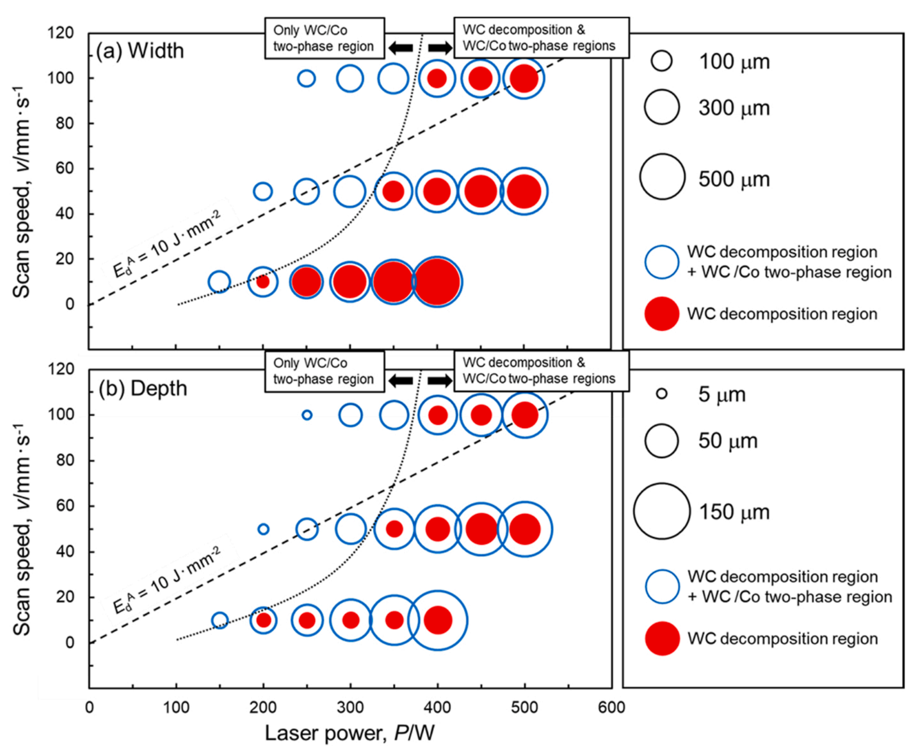

Fig. 5. Bubble chart of (a) width and (b) depth of WC decomposition region and WC decomposition region + WC/Co two-phase region plotted as functions of laser power and scan speed.

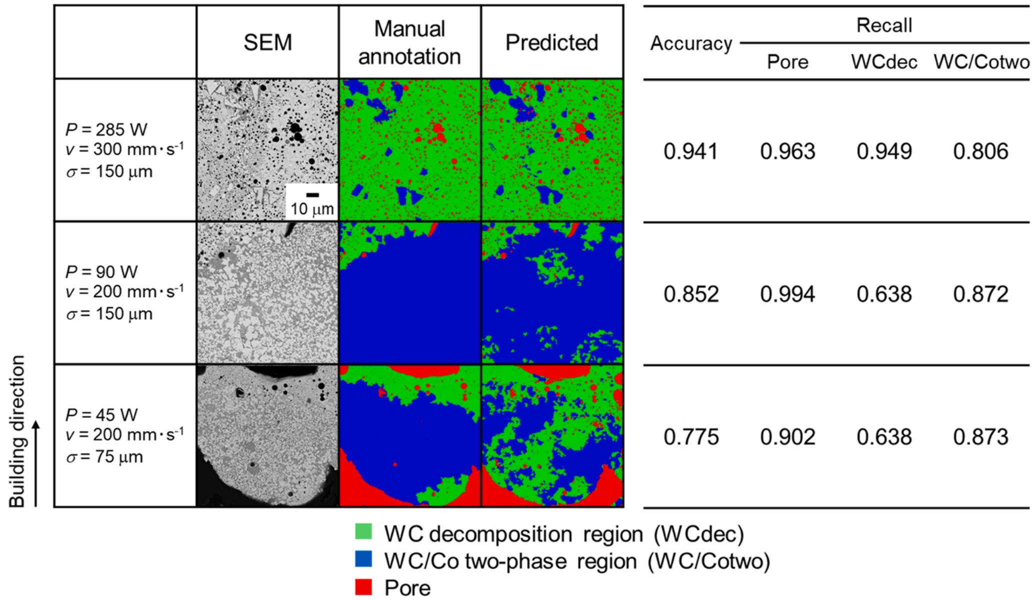

Fig. 6. SEM, manually annotated, and CNN annotated images and accuracy and recall values of laser powder bed fused WC/Co cemented carbides fabricated under various laser conditions when only Fig. 2(a) was trained to the CNN.

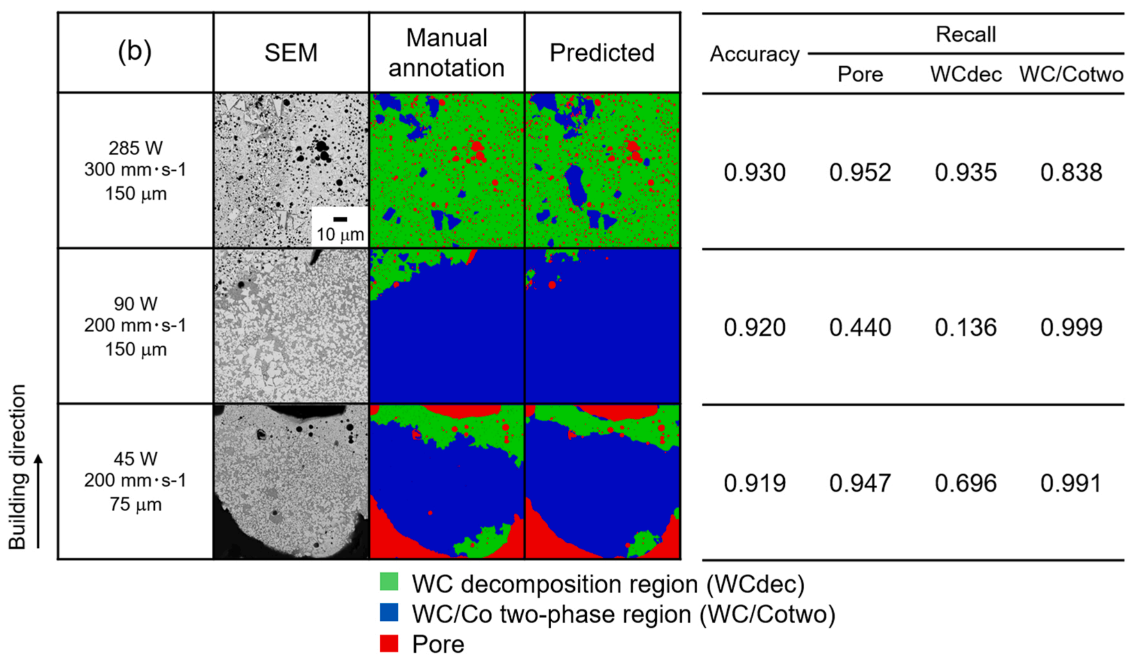

Fig. 7. SEM, manually annotated, and CNN annotated images with accuracy and recall values of laser powder bed fused WC/Co composites fabricated under various laser conditions when both Fig. 2(a) and (b) were trained to the CNN.

CNN annotated images of L-PBF-fabricated WC/Co composites fabricated under various conditions. Also, the change in area fraction of WC decomposition region with area energy density (E_d^A) (Fig. 8(c)) and linear sum of standardized laser power, scan speed, and spot size weighed by linear regression coefficient (Fig. 8(d)) are presented. In these figures, the laser conditions under which the samples were unconsolidated are denoted by cross symbols. In Fig. 8(c, d), A_d in the unconsolidated samples is plotted at zero. The value of A_d decreased with a decrease in laser power and an increase in scan speed (Fig. 8(a)). In addition, when almost the same laser power and scan speed were applied, A_d decreased with an increase in spot size (Fig. 8(a), (b-1), and (b-3)). When A_d was plotted as a function of E_d^A, A_d tends to increase with an increase in E_d^A. However, at E_d^A of approximately 4.5 J•mm^{-2}, A_d values were distributed from nearly 0–1. The large variation indicates that the contributions of P, v, and σ were not considered appropriately by E_d^A. When the relationship between A_d and laser conditions were analyzed by MLR, the regression equation, $A'_d = 1.12\,P' - 0.52\,v' - 0.74\sigma'$ was obtained. The probability (P) value of this equation was 8.43×10^{-5}, which was lower than Bonferroni's corrected significance level ($0.05/3 \approx 0.017$), indicating the statistical high reliability of this equation. The significance level of standardized partial regression coefficients of laser power (1.12) and spot size (−0.74) were 0.0017 and 0.0041, respectively, which was lower than 0.05. By contrast, the significance level of the standardized partial regression coefficient of scan speed (−0.52) was 0.0779, indicating a slightly low reliability of this coefficient. The equation indicates that A_d increased with an increase in

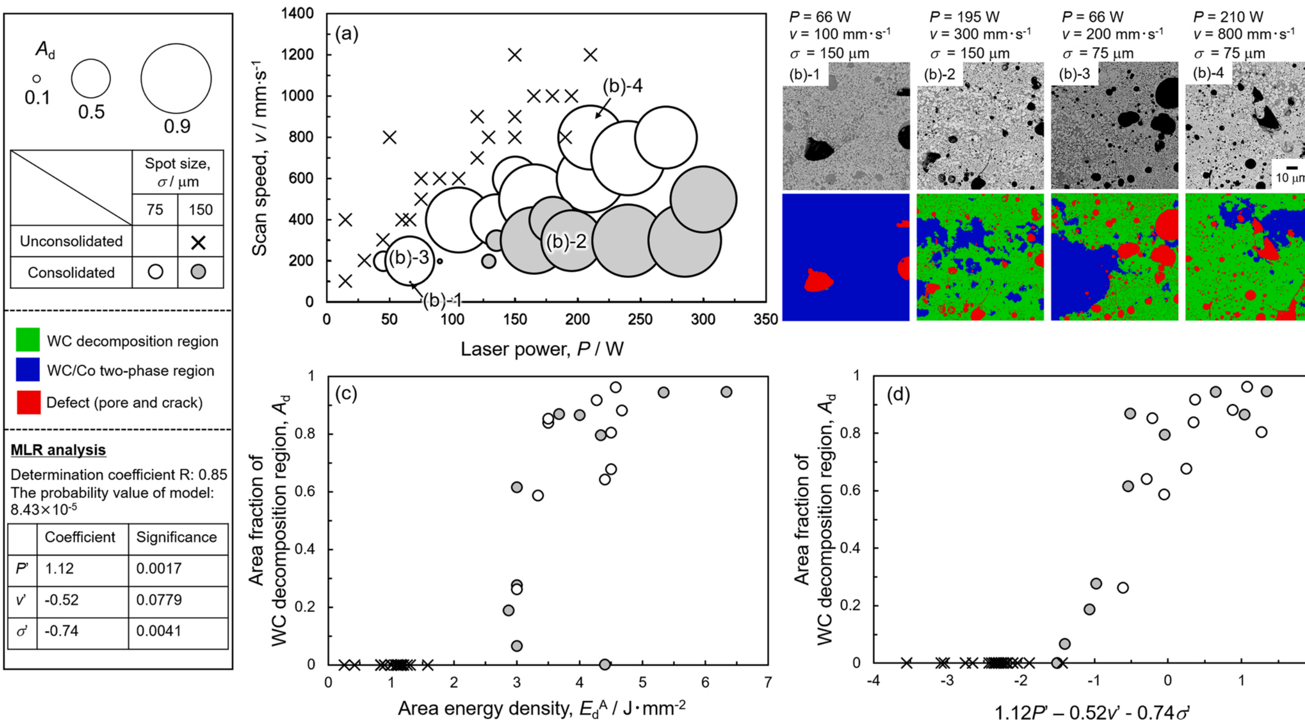

Fig. 8. (a) Bubble chart of area fraction of WC decomposition region plotted as functions of laser power and scan speed. (b) Representative SEM and CNN annotated images of laser powder bed fused WC/Co composites. Changes in area fraction of WC decomposition region with (c) area energy density, and (d) linear sum of standardized laser power (P'), scan speed (v'), and spot size (σ') weighed by linear regression coefficient.

laser power and a decrease in scan speed and spot size. The absolute value of the standardized partial regression coefficient represents the contribution of each laser parameter. Among all the parameters, the laser power had the largest contribution, and the scan speed had the smallest contribution. The larger contribution of laser power compared with scan speed was consistent with the results of single-laser-scan experiment (Fig. 5). As shown in Fig. 8(d), A_d increased with increasing 1.12 P' – 0.52 v' – 0.74σ'. When the 1.12 P' – 0.52 v' – 0.74σ' was below approximately – 1.5, A_d became almost zero. However, the cross symbols representing unconsolidated conditions were also plotted around 1.12 P' – 0.52 v' – 0.74σ' of – 1.5. Decreasing 1.12 P' – 0.52 v' – 0.74σ' to shrink the WC decomposition region leads to unconsolidation of the sample. Therefore, the laser conditions consolidating the sample and suppressing the formation of WC decomposition region were explored by SVM.

Fig. 9 presents the process parameter classification by support vector machine. The classification based on consolidated/unconsolidated (Fig. 9(a,c)) and $A_d < 0.1/\geqq 0.1$ (Fig. 9(b,d)) is shown. All data were classified appropriately using hyperplanes. The planes on σ = 150 µm in Fig. 9(a, b) are shown in Fig. 9(c, d). The consolidated/unconsolidated boundary is the line rising to the right, indicating that both the laser power and scan speed contributed to the consolidation of the samples. The line passes near the origin which indicate that the line is located under conditions of same energy density. It is known that the relative density of L-PBF-fabricated WC/Co composite correlates linearly with the volumetric energy density [24]. The boundary in Fig. 9(c) is reasonable considering the decrease in relative density with a decrease in energy density; therefore, results in non-consolidation of the sample. By contrast, the boundary of $A_d < 0.1/\geqq 0.1$ is a vertical line, which indicates that the scan speed scarcely contributed to the formation of the WC decomposition region. This tendency was consistent with the MLR analysis. Therefore, the contributions of the laser parameters to consolidation were different from those to the formation of WC decomposition region.

An SVM-based process map constructed by superimposing Fig. 9(c, d) is shown in Fig. 10 (a). It is expected that the consolidated sample with $A_d < 0.1$ can be manufactured under the laser conditions marked by bold triangle in Fig. 10 (a). The samples were fabricated under the recommended conditions as well, and evaluated by the microstructural observations and CNN semantic segmentation. The SEM and corresponding CNN-annotated images of the samples fabricated under the recommended conditions are shown in Fig. 10 (b). For comparison, a sample was also fabricated under a condition marked outside the bold triangle in Fig. 10 (a), and the images are shown in Fig. 10 (c). The A_d values were also shown in Fig. 10 (b, c). Under all recommended conditions, the samples were consolidated and had an A_d below 0.1. The consolidation and suppression of WC decomposition was achieved under SVM recommended conditions. By contrast, under non-recommended conditions, A_d exceeded 0.1, as shown in Fig. 10 (c). Therefore, it is demonstrated that the SVM-based process map is useful for filtering laser conditions.

Fig. 11 shows the low-magnification SEM images showing the pores and defects of the laser powder bed fused WC/Co composites fabricated under the SVM recommended conditions shown in Fig. 10 (a). The relative density obtained by the image-analysis using the SEM images were also shown in the figure. The specimens became denser as the laser power increased and the scan speed decreased. The maximum relative density obtained in this study was 98.5%. However, in the densest specimen, the pores and cracks has been observed. To eliminate these defects and to evaluate the mechanical properties, the heat treatment and HIP treatment were carried out.

Fig. 12 (a) shows the Vickers hardness and transverse rupture strength of WC/Co composites fabricated by L-PBF under SVM-recommended ((P, v, σ) = (66 W, 100 mm•s^{-1}, and 150 µm)) and non-recommended conditions ((P, v, σ) = (195 W, 300 mm•s^{-1}, and 150 µm)). These samples were heat treated at 1380 °C for 4 h and hot isostatic pressed at 1350 °C for 4 h under 100 MPa prior to mechanical testing. For comparison, the mechanical properties of conventionally sintered WC/Co composites [34–43] are also shown in Fig. 12 (a). Both the Vickers hardness and transverse rupture strength of the samples

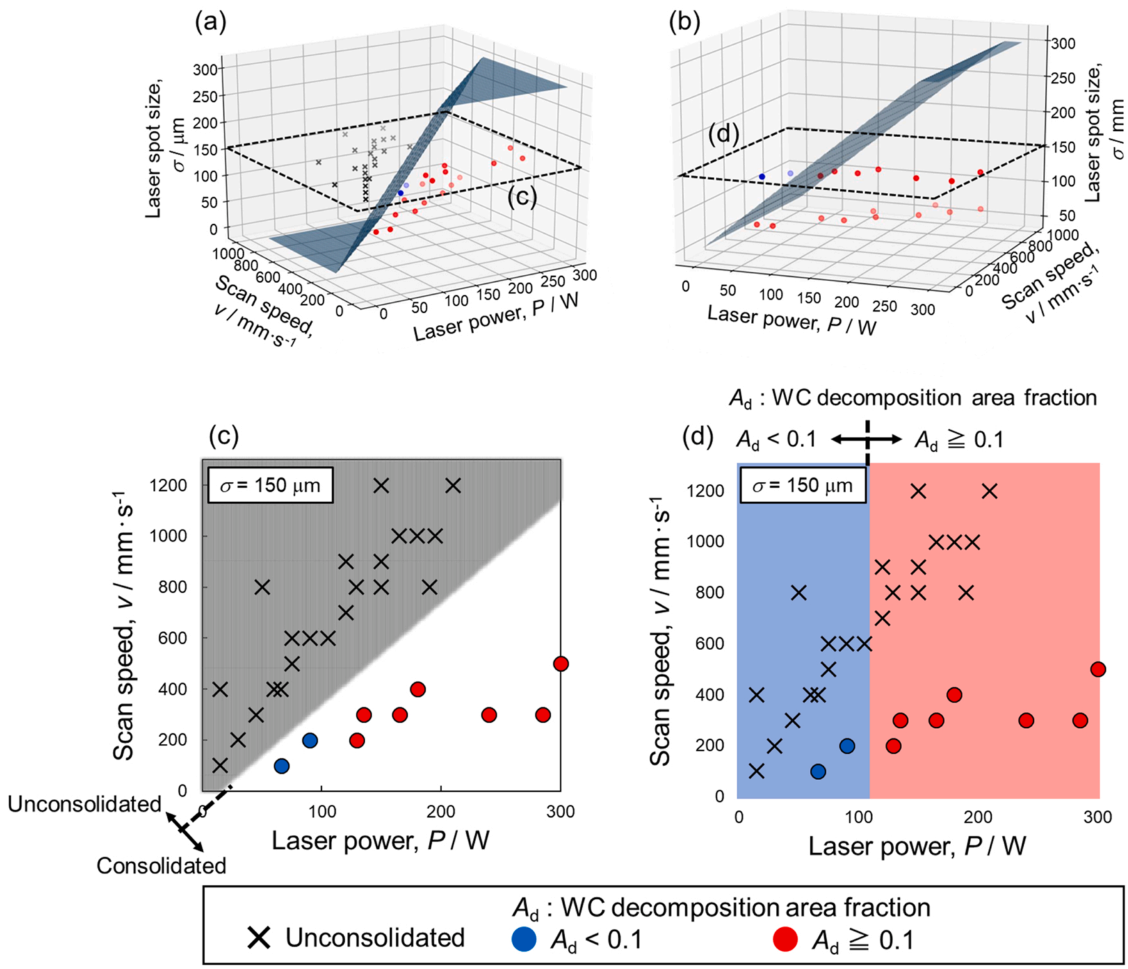

Fig. 9. Process parameter classification by support vector machine: classification based on (a, c) consolidated/unconsolidated and (b, d) $A_d < 0.1/\geqq 0.1$. The planes on $\sigma = 150$ μm in (a, b) are displayed in (c, d).

fabricated under the recommended condition (☆) were higher than those of the samples fabricated under the non-recommended condition (□). The microstructures of the tested samples are shown in Fig. 12 (b, c). Both the samples were composed of a WC/Co two-phase microstructure. The Co and C content of raw powder and the as-built specimens are listed in Table 2. Both as-built specimens had almost the same C content as the raw powder. However, the Co content of the specimen fabricated under the non-recommended condition was obviously lower than that of raw powder. The specimen fabricated under the SVM-recommended conditions almost maintained the Co content of the raw powder. Although the composition was shifted in the specimen fabricated under the non-recommended condition, the W_2C and W_3Co_3C phases were not stable at the heat treatment temperature [21]. The samples fabricated under the recommended and non-recommended conditions had WC particles with sizes of approximately 1 and 10 μm, respectively. Thus, after the high-temperature treatments, the samples fabricated under the recommended conditions had a smaller WC particle size compared with those fabricated under non-recommended conditions. It is well known that the hardness and strength are enhanced by refining the WC particle size. Refinement is a dominant contributor to the enhanced Vickers hardness and transverse rupture strength. The enhanced Vickers hardness and transverse rupture strength reached the class of conventional sintered WC/Co composites but they were at a low level. Further optimizations, including heat treatment conditions and other laser powder bed fusion process parameters, are required.

4. Discussion

In this study, the effects of laser power (P), scan speed (v), and spot size (σ) on the microstructure of L-PBF-fabricated WC/Co composites were investigated. The single-laser scan experiment revealed that the laser power contributed significantly to the formation of the WC decomposition region than scan speed (Figs. 4 and 5). The MLR analysis of the L-PBF-fabricated WC/Co-composites demonstrated that the laser power contributed largely to the increase in the WC decomposition region (Fig. 8). The spot size also contributed to the increase in the WC decomposition region compared to the scan speed (Fig. 8). This tendency was also confirmed by SVM analysis (Fig. 9). The differences in the contributions of each laser parameter are discussed.

A previous study [24] and the single-laser scan results from the present study revealed that the WC decomposition region was formed closer to the laser-irradiated spot (Fig. 3(b)). The WC decomposition region was formed by melting the WC phase at high temperatures. By contrast, the WC/Co two-phase region was formed far from the laser-irradiated spot. The microstructural characteristics indicated that only the Co phase was melted by heat transfer from the laser-irradiated surface (Fig. 3(c) and 4 (f)). The melting temperatures of WC and Co are 2870 and 1520 °C, respectively. The large difference in the melting temperatures results in the formation of two regions with different microstructural characteristics. In the region where the maximum temperature exceeds the melting temperature of WC, both WC and Co melt to form the WC decomposition region. In the region where the maximum temperature is between the melting temperatures of WC and

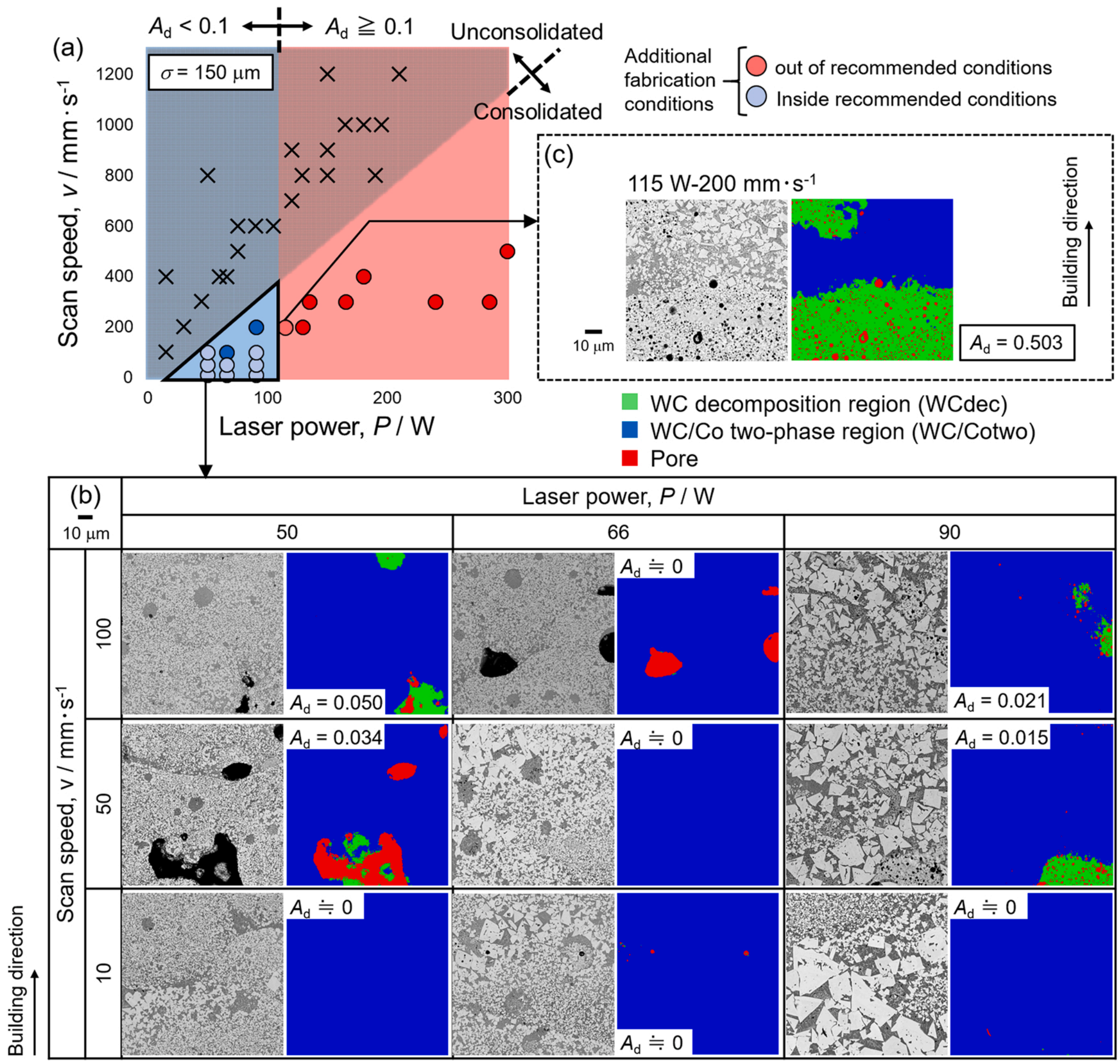

Fig. 10. (a) SVM-based process map obtained by overlapping Fig. 9(c, d). (b, c) SEM and corresponding CNN annotated images of laser powder bed fused WC/Co composites fabricated under laser conditions (b) recommended and (c) non-recommended by SVM.

Co, only Co melt, and subsequent solidification results in the formation of WC/Co two-phase region. The laser parameters contribute to the maximum temperature during the L-PBF process and influence the formation and increase of WC decomposition region. Therefore, the maximum temperature is important for the formation of WC decomposition region.

To understand the effect of laser parameters on the maximum temperature, the moving heat source model [44] was applied. Fig. 13 (a) shows a schematic of the moving heat source model. It is assumed that the heat source has the following Gaussian energy distribution:

$$Q = \frac{8P}{\pi\sigma^2}\exp\left[-\frac{8\left\{(x-v\tau)^2+y^2\right\}}{\sigma^2}\right], \tag{6}$$

where Q is the heat input per unit area and τ is the time. The heat source moves in the x-direction from the origin at a constant speed v (scan speed). z is the direction perpendicular to the laser-scanned surface and y is the direction perpendicular to the x- and z-directions. The temperature field (T (x, y, z, τ)) can be calculated using the following equation:

$$T\left(x,y,z,\tau\right) - T_0 = \frac{P}{\pi\rho c\sqrt{4\pi\alpha}}\int_0^{\tau}\frac{\exp\left(-\frac{(x-v\tau')+y^2}{4\alpha\left(\tau-\tau'\right)+2(\sigma/4)^2}-\frac{z^2}{4\alpha(\tau-\tau')}\right)}{\sqrt{\tau-\tau'}\left\{2\alpha\left(\tau-\tau'\right)+(\sigma/4)^2\right\}}\mathrm{d}\tau', \tag{7}$$

where T_0 is the ambient temperature, ρ is the density, c is the specific heat, and α is thermal diffusivity. Although the Eq. (7) does not include the melting and evaporation of the material, liquid flow in the molten pool, the existence of the powder layer, and laser absorptivity of the material, the contributions of laser power, scan speed, and spot size to the increase in temperature can be understood in general. The integral term in Eq. (7) was numerically calculated using the scipy of a Python library [45]. Fig. 13 (b) presents the time dependence of the temperature at the surface of laser-irradiated WC/Co composites. To calculate the temperature, ρ, c, and α are set at 1.3×10^4 kg•m^{-3}, 262 J·kg^{-1}·K^{-1}, and 4.4×10^{-5}, m^2/s, respectively [46,47]. Here, one condition of (P, v, σ) = (150 W, 100 mm·s^{-1}, 150 μm) is set as the base condition (i). From the base condition (i), the following conditions are applied: (ii) scan speed is reduced to half of the base condition (i), (iii) the spot size is decreased to half of the base condition (i), and (iv) the laser power is doubled as the base condition (i). The condition (ii)–(iv) have the same area energy density as expressed in Eq. (5). The temperature shown in Fig. 13 (b) is normalized by the peak temperature of the base condition (i) to compare the contribution of each laser parameter to the maximum temperature. The target coordinates are (x_0, 0, 0), which indicate the center of the laser-scanning track. When $\tau - x_0/v$ of the horizontal axis is 0, the center of the laser spot is located at the target coordinate and the temperature reaches its maximum. When the scan speed is halved from

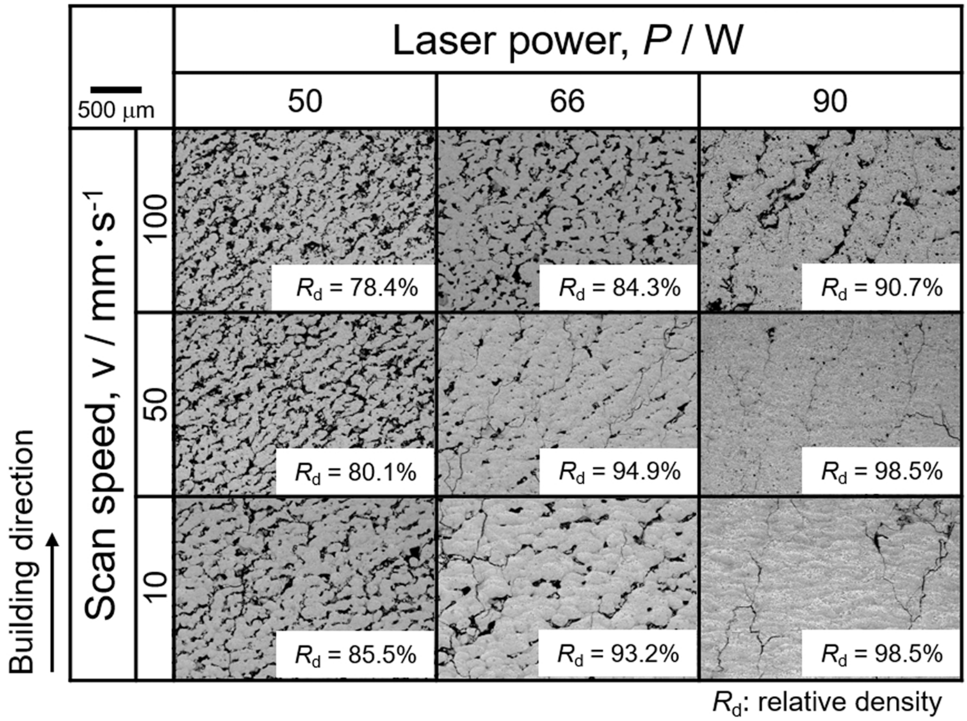

Fig. 11. SEM images showing pores and cracks in the WC/Co composites fabricated by L-PBF under the SVM-recommended conditions shown in Fig. 10. The relative density (R_d) obtained by the image analysis was also shown in the figure.

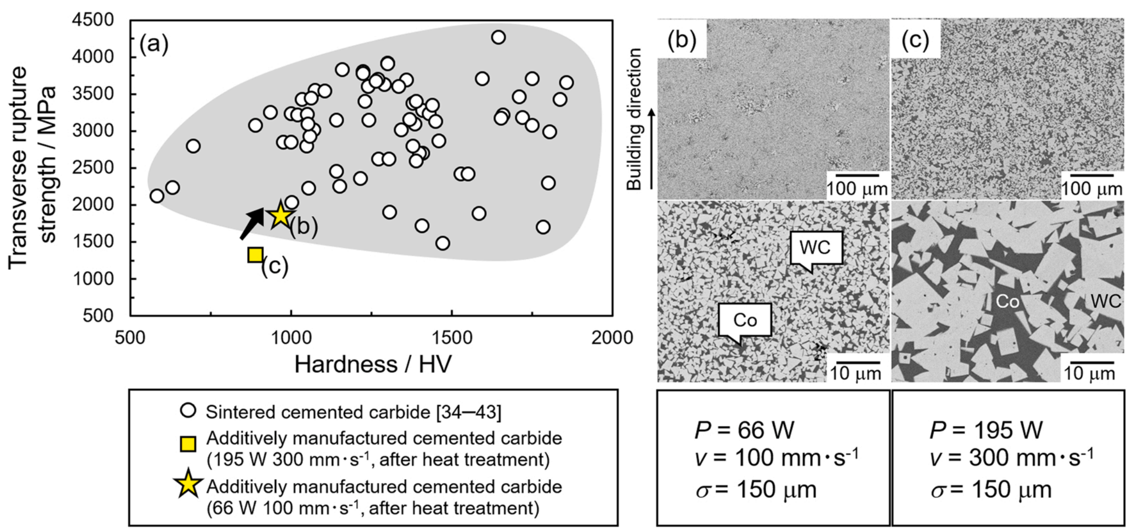

Fig. 12. (a) Transverse rupture strength and Vickers hardness. (b, c) Low and high magnification SEM images of laser powder bed fused WC/Co composites heat-treated at 1380 ℃ for 4 h and hot isostatic pressed at 1350 ℃ for 4 h under a pressure of 100 MPa, respectively. For comparison, the transverse rupture strength and Vickers hardness of conventional sintered WC/Co composites were also shown (Fig. 12 (a)).

Table 2
Chemical composition of the raw powder and as-built samples fabricated under the SVM-recommended and non-recommended conditions.

	Composition (mass%)		
	W	Co	C
Raw powder[21]	Bal.	25.63	4.59
As-built sample (SVM-recommended condition) (P, v, σ) = (66 W, 100 mm·s^{-1}, 150 μm)	Bal.	25.35	4.68
As-built sample (Non-recommended condition) (P, v, σ) = (195 W, 300 mm·s^{-1}, 150 μm)	Bal.	24.33	4.56

the base condition (ii), the maximum temperature barely changes, although the peak corresponding to temperature is broadened. This is because the contribution of scan speed in Eq. (7) is limited to the exponential term, which represents the temperature distribution. If the reaching temperature does not exceed the melting temperature of WC at the base condition (i), then the reaching temperature is below the melting temperature of WC under the slow scan speed condition (ii) as well. This is in good agreement with the smallest contribution of scan speed to the formation and increase of the WC decomposition region. When the spot size is halved or the laser power is doubled (condition (iii) or condition (iv)) from the base condition (i), the maximum temperature achieved is approximately doubled. These two parameters contribute to

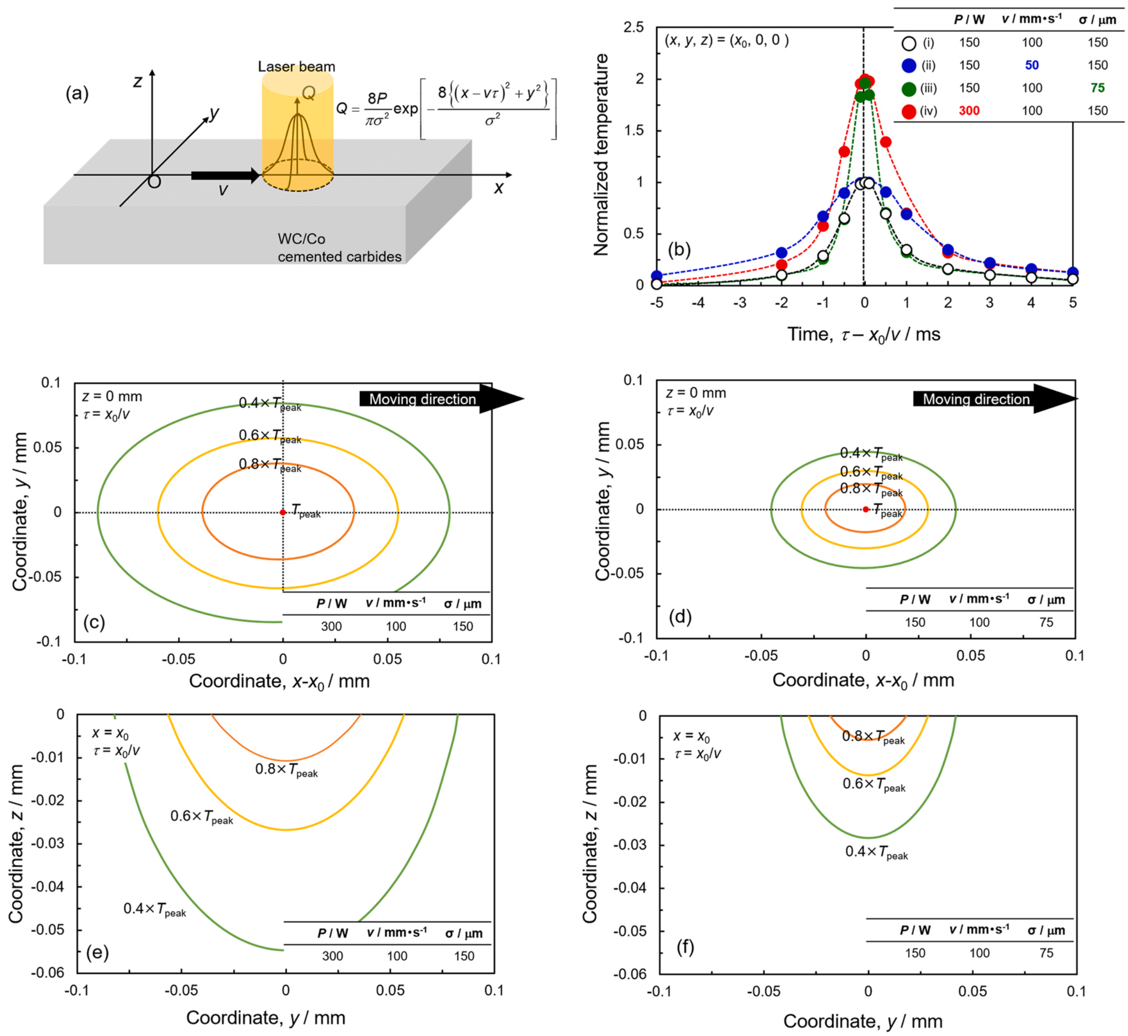

Fig. 13. (a) Schematic of moving heat source model with Gaussian energy distribution and (b) changes in normalized temperature at a center of laser spot ((x_0, 0, 0)) with respect to time under various laser conditions. The temperature was normalized by the peak temperature under a laser power (P) of 150 W, a scan speed of (v) 100 mm·s^{-1}, and a spot size (σ) of 150 µm. (c–f) Contour map of temperature at $\tau = x_0/v$ on sections at (c, d) $z = 0$ mm and (e, f) $x = x_0$ under the laser conditions of (c, e) (P, v, σ) = (300 W, 100 mm·s^{-1}, 150 µm) and (d, f) (150 W, 100 mm·s^{-1}, 75 µm), respectively.

the maximum temperature more significantly than the scan speed, which is in accordance with the larger contributions of laser power and scan speed to the area fraction of WC decomposition region. The contribution of laser power is understood as laser power is the only coefficient of the integral term in Eq. (7). The spot size is not only in the exponential term, but also present as the denominator in the integral term, which can generate a larger contribution than the scan speed. Although the maximum temperature under conditions (iii) and (iv) are almost the same, the peak corresponding to the temperature at condition (iii) is sharper compared to that of condition (iv). It indicates that only a narrow region is irradiated upon using a smaller spot size laser. Fig. 13 (c, d) shows the temperature distribution at $z = 0$ and $\tau_0 = x_0/v$ under conditions (iii) and (iv). In the figures, the position of (x-$v\tau_0$, y) = (0, 0) is the center of the laser spot, and the temperature is the peak temperature (T_{peak}), as shown in Fig. 13 (b). The ellipses in the figures indicate isotherm lines of 0.8, 0.6, and 0.4 times of T_{peak}. The temperature distribution under condition (iv) is wider than that under condition (iii), indicating that the temperature under condition (iv) increases to a higher temperature over a wider range than that under condition (iii). The tendency is also confirmed on the y-z plane shown in Fig. 13 (d, f). This causes a larger contribution of the laser power to the area fraction of the WC decomposition region than that of the spot size. The above discussion is schematically summarized in Fig. 14. Although a decrease in the scan speed enlarges the molten pool, the peak temperature changes only to a small extent, resulting in an insignificant contribution to the formation of the WC decomposition region. When the spot size is decreased or laser power is increased, the peak temperature increases significantly, contributing to the formation and enlargement of the WC decomposition region. In particular, an increase in laser power increases the temperature over a wider range, which contributes the most to the increase of the WC decomposition region.

The discussion based on the moving heat source model also suggest the physical validity of the contribution order of laser power, spot size, and scan speed to the WC decomposition. In this study, changes in the processing parameters are imbalanced as shown in Fig. 1(b). There is a possibility that the imbalance would affect the results of data science approaches and provide incorrect contributions of laser parameters to the WC decomposition. The moving heat source model is a physical model for predicting the temperature of laser irradiated parts. The fact that the laser power increases the temperature the most significantly in the widest range would verify the largest contributions of laser power to the WC decomposition region obtained by the data science approaches. However, to quantitatively understand the contributions of laser parameters, in situ temperature measurements and thermal fluid analysis will be required in future studies.

A lower laser power, lower scan speed, and larger spot size would be

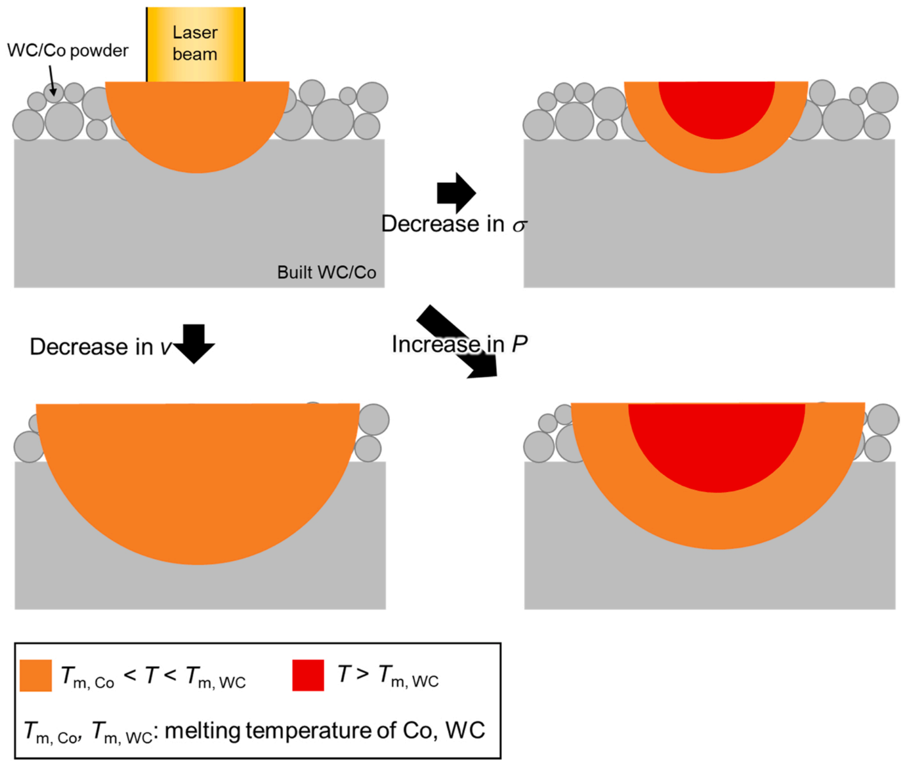

Fig. 14. Schematic illustration of the contributions of laser power (P), scan speed (v), and spot size (σ) to the temperature field and formation of the WC decomposition region.

preferable for the L-PBF of other composite materials. Many composites, including Al/TiB_2, Ni/TiB_2, and Al/carbon fiber [48–50] have reinforcements with higher melting temperatures than the matrices. The microstructure changes with respect to the melting process of reinforcements and two types of microstructures (corresponding to WC decomposition and WC/Co two-phase regions) will be formed. To prevent the melting of reinforcements in composite materials, a low laser power, low scan speed, and large spot size are preferable.

5. Conclusion

In the present study, single-laser scan of sintered WC/Co composites was carried out to explore the laser conditions on suppressing WC decomposition. Using a CNN model to which the micrographs of the single-laser-scanned WC/Co composites and low-magnification micrographs of the sintered WC/Co composites were trained, the area fractions of the WC decomposition region in the L-PBF-fabricated WC/Co composites were quantified. The contributions of laser parameters to the area fraction of the WC decomposition region were analyzed by multiple linear regression (MLR) and numerical calculations based on the moving heat source model. The laser conditions for suppressing the WC decomposition region were explored using a support vector machine (SVM). The main findings are summarized as follows.

1) Laser irradiation of the sintered WC/Co-composite generates two different microstructural regions. One region is similar to conventional WC/Co composites and was composed of WC and Co phases (WC/Co two-phase region). The other region includes pores, W_2C, and W_3Co_3C phases (WC decomposition region).
2) The CNN model to which only the micrographs of the single-laser-scanned WC/Co composites were trained could not segment the microstructural region in the L-PBF-fabricated WC/Co composites correctly because the single-laser-scanned WC/Co composites (sintered WC/Co composites) have larger WC particles in the WC/Co two-phase region. The accuracy can be improved by training low-magnification micrographs of sintered WC/Co composites to the CNN. For the powder bed fused materials, not only the images of built specimens but also those of single-laser-scanned specimens is useful for training the CNN to classify microstructures.
3) An increase in the laser power contributes the most to increasing the WC decomposition region, whereas the decrease in scan speed results in an insignificant contribution. An increase in the laser power increases the temperature over a wide range, leading to the formation and enlargement of the WC decomposition region. A decrease in scan speed does not contribute to the maximum temperature, resulting in an insignificant contribution toward increasing the WC decomposition region.
4) The SVM-based process map created using the CNN-annotated results provides the laser conditions for fabricating consolidated specimens while suppressing the formation of the WC decomposition region. Under the recommended conditions, samples with small WC decomposition regions has been successfully fabricated.

CRediT authorship contribution statement

Asuka Suzuki: Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Funding acquisition, Writing - original draft. **Yusuke Shiba:** Methodology, Investigation, Formal analysis, Data curation. **Hiroyuki Ibe:** Resources, Investigation, Data curation, Formal analysis, Writing - review & editing. **Naoki Takata:** Writing – review & editing, Validation, Resources, Methodology. **Makoto Kobashi:** Writing – review & editing, Validation, Resources, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgements

The present work is supported by the Priority Research Project of Aichi Prefectural Government, “Knowledge Hub Aichi” and JSPS KAKENHI Grant Number JP22H05281 (Grant-in-Aid for Transformative Research areas).

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